

# Verdant-Minds: A Hybrid Cognitive Architecture with Graph Memory, Wave-State Dynamics, and Measurable Emergent Concept Scaffolding

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## Abstract

Verdant-Minds is a research-grade cognitive architecture that processes text through a structured nine-stage pipeline (sensory  $\rightarrow$  pattern  $\rightarrow$  memory  $\rightarrow$  communication  $\rightarrow$  reasoning  $\rightarrow$  ethics  $\rightarrow$  action  $\rightarrow$  language  $\rightarrow$  learning). The system maintains a symbolic MemoryWeb (graph memory) coupled bidirectionally to an Extended Cognitive Wave Function (ECWF), a compact, configurable wave-state model whose entropy, magnitude, and phase influence memory reinforcement, decay, and concept formation. A governance layer (“Three Kings”) provides oversight across data quality, ethics, and action selection.

We report evidence from cultivation runs that emergent concepts—concepts created autonomously by the system’s wave dynamics rather than seeded by designers—not only appear, but preferentially link to earlier emergent concepts in the memory backbone. In a representative run (119 concepts, 37 emergent nodes), emergent $\rightarrow$ emergent edges in a sparsified backbone exhibit a strong timing bias toward linking newer emergents to older emergents (observed earlier-share 0.807; shuffle baseline  $\sim 0.499 \pm 0.083$ ;  $z \approx 3.71$ ), suggesting temporal scaffolding during concept formation. These results are not claims of consciousness; they are measurable structural properties of an evolving concept graph coupled to a state-dynamics core.

**Keywords:** cognitive architecture, emergent concepts, graph memory, wave-state dynamics, ethomorphism, temporal scaffolding

## 1 Motivation

Modern AI systems are dominated by large neural models that are powerful but opaque. Verdant-Minds explores an alternate route: a modular cognitive pipeline with explicit memory, governance, and measurable internal state dynamics. The goal is not to replace large language models but to create an architecture where:

- memory is explicit and inspectable,
- ethics and action gating are first-class processing stages,
- internal state evolution can be instrumented and measured,
- and emergent structure can be observed and quantified over time.

The project is developed under the banner of *Ethomorphic AI*—the principle that ethical reasoning is geometrically embedded in cognitive state space rather than applied as an external constraint. Verdant-Minds is the first implementation of this principle.

## 2 System Overview

Verdant-Minds is organized as five interacting subsystems:

1. **Nine-block cognitive pipeline:** deterministic sequencing of processing stages from sensory input through continual learning.
2. **MemoryWeb:** graph-based concept memory with reinforcement, decay, and spreading activation retrieval.
3. **ECWF core:** multi-facet complex-valued wave state representing cognitive and ethical dimensions; produces entropy, magnitude, and phase signals.
4. **Memory↔ECWF Bridge:** bidirectional coupling where memory influences wave parameters and wave outputs reinforce memory and generate new concepts.
5. **Three Kings governance:** oversight and conflict resolution at key points in the pipeline.

This architecture is designed to be inspected, instrumented, and adapted—closer in lineage to cognitive architectures such as SOAR [Laird(2012)], ACT-R [Anderson et al.(2004)], and OpenCog [Goertzel et al.(2014)] than to chat-style applications.

## 3 Architecture Details

### 3.1 Nine-Block Pipeline

Each input is wrapped into a shared state container (CognitiveChunk) that accumulates sections and a processing log. Blocks read prior sections and write their own, yielding a traceable per-input “cognitive episode.”

The processing order is: Sensory Input → Pattern Recognition → Memory Storage → Internal Communication → Reasoning & Planning → Ethics & Values → Action Selection → Language Processing → Continual Learning. The final block may trigger emergent concept detection.

### 3.2 MemoryWeb (Graph Memory)

MemoryWeb is a concept graph implemented over NetworkX in which nodes store stability scores, access counts, timestamps, metadata, and weighted connection lists. Edges may be gated by an ethical-distance-dependent probability (“pconnect”):

$$p_{\text{accept}} = (1 - p_{\text{eth}} \cdot \delta_e) \cdot \exp(-\beta_E \cdot \Delta_E), \quad p_{\text{accept}} \in [0, 1] \quad (1)$$

where  $p_{\text{eth}} = 0.6$  and  $\beta_E = 1.0$  are default constants,  $\delta_e$  is a local ethical distance, and  $\Delta_E$  is a global ethical distance measure. An edge is accepted if a uniform random draw falls below  $p_{\text{accept}}$ .

### 3.3 ECWF (Wave-State Core)

The Extended Cognitive Wave Function represents system state as a sum of complex facets:

$$\Psi(x, e, t) = \sum_{i=1}^n M_i(x, e, t) \cdot \exp(j(2\pi k_i \cdot x + 2\pi m_i \cdot e - \omega_i t + \phi_i)) \quad (2)$$

where  $x \in \mathbb{R}^{d_c}$  are cognitive dimensions,  $e \in \mathbb{R}^{d_e}$  are ethical dimensions,  $t$  is time,  $M_i$  is the amplitude function for facet  $i$ ,  $k_i$  and  $m_i$  are wave numbers,  $\omega_i$  is angular frequency, and  $\phi_i$  is phase shift. The system computes Shannon entropy over  $|\Psi|^2$  as an uncertainty signal, and numerical sensitivity gradients via finite-difference perturbation.

ECWF also maintains short-horizon history coupling through exponentially decayed averaging of past states, enabling dynamic dependence on recent cognitive history.

### 3.4 Memory↔ECWF Bridge

The bridge performs two-way coupling each processing cycle:

- **Memory** → **Wave**: active concepts modulate ECWF parameters via influence vectors derived from concept stability and dimension mappings.
- **Wave** → **Memory**: wave outputs activate concepts through dimension reverse-lookup, reinforce or decay stability based on thermodynamic phase, and detect conditions for emergent concept creation.

Concept-dimension mappings are initialized using sentence-transformer embeddings (all-MiniLM-L6-v2) projected via PCA to the ECWF dimension space, ensuring that semantically related concepts map to nearby regions in the wave function.

### 3.5 Governance (“Three Kings”)

Governance is layered oversight operating at specific pipeline stages:

- **Data King**: information quality assessment and anomaly detection (operates at Block 4).
- **Ethics King**: ethical evaluation and response modulation, incorporating coherence tension signals (operates at Block 6).
- **Forefront King**: phase-aware attention allocation, capacity controls, and decision threshold management (operates at Block 7).

The Three Kings Council coordinates conflict resolution for critical decisions via weighted influence and consensus mechanisms.

### 3.6 Thermodynamic Phase Control

The system employs a glass-transition temperature parameter ( $T_g$ ) as a first-class control signal that modulates processing behavior:

- **Rigid** ( $T_g < 0.4$ ): low decay, low reinforcement, high decision thresholds—stabilization behavior.
- **Flexible** ( $0.4 \leq T_g \leq 0.6$ ): balanced exploration/exploitation.
- **Chaotic** ( $T_g > 0.6$ ): high decay, high reinforcement, low decision thresholds—exploratory behavior.

### 3.7 Coherence Geometry

Each cycle computes coherence invariants from wave, ethics, and memory telemetry. The system constructs a generalized metric  $\tau_\alpha(a, b) = |a - b|^\alpha$  and checks triangle inequality validity across an  $\alpha$ -grid. The Housed Contradiction Index (HCI) quantifies internal cognitive tension. These signals feed back into governance: the Forefront King adjusts its decision threshold under coherence strain, and the Ethics King modulates its overall score when geometric tension is detected.

## 4 Emergent Concept Formation

### 4.1 Trigger Conditions

Emergent concept creation is gated by multiple conditions that must all be satisfied simultaneously:

1. **Magnitude gate**: wave magnitude must exceed a minimum threshold (default 0.15) at the callsite in the continual learning block.
2. **Entropy window**: sensitivity entropy must fall within  $[0.3, 3.0]$ , rejecting states that are too ordered or too chaotic for meaningful pattern formation.

3. **Strength gate:** mean sensitivity magnitude must exceed the detector threshold (default 0.15).
4. **Novelty gate:** the combination key (sorted top-3 concept identifiers) must not have been seen before.

When semantic embeddings are available, candidate ranking blends stability with inverse semantic similarity (“surprise”). This biases the system toward creating concepts that bridge semantically distant parents—non-obvious combinations that pure graph proximity would not surface.

## 4.2 Creation and Integration

New emergent concepts are added to the MemoryWeb with structured metadata (origin, creation time, entropy, magnitude) and connected to their parent concepts. Critically, emergent concepts receive dimension mappings in the ECWF space, making them visible to future wave dynamics. This means emergent concepts can participate in the activation of subsequent emergent concepts, enabling second-order associative structure.

# 5 Experimental Setup

## 5.1 Cultivation Loop

An external LLM provider (Groq, Anthropic, or Mistral) acts as a cultivation tutor, generating inputs designed to produce cognitive tension in the system. The cultivator monitors HCI, phase state, and emergent concept generation. It employs a domain rotation wheel (9 domains), phase-aware input selection (rigid→depth, flexible→paradox, chaotic→grounding), and forced perturbation at configurable intervals to prevent phase plateaus.

Each cycle, the system processes the input through the full nine-block pipeline, updates memory and wave state, and logs structured telemetry to a JSONL cycle log.

## 5.2 Instrumentation

For each cycle we capture: MemoryWeb size and topology, emergent concept count and identities, coherence invariants (HCI, triangle validity, violation rate,  $\alpha$ -critical), thermodynamic phase state, and per-block processing metrics. Full system state is persisted as JSON at configurable intervals.

## 5.3 Representative Run Parameters

The results reported below are drawn from a representative cultivation run with the following parameters:

- Seed: 42 (NumPy RNG + ECWF random state)
- Cycles: 91
- Provider: Groq (llama-3.3-70b-versatile)
- Perturbation interval: 15 cycles
- ECWF dimensions: 5 cognitive + 5 ethical
- Initial knowledge base: 82 seeded concepts across 9 domains

# 6 Results: Temporal Scaffolding in Emergent Concept Links

## 6.1 Run Summary

At the conclusion of the representative run:

- Total concepts: 119 (82 seeded + 37 emergent)

- MemoryWeb backbone: 513 edges (top-6 edges per node sparsification)
- Self-cluster average access count: 2895.2
- Non-self-cluster average access count: 8.4
- Access count ratio: 345×

## 6.2 Key Finding: Preferential Attachment to Earlier Emergent Concepts

We analyzed emergent→emergent (EE) edges in the sparsified backbone and oriented each edge by creation order. Ordering is determined primarily from the `creation_time` field stored in each emergent concept’s metadata (a float timestamp written at concept birth); where metadata is unavailable, the integer timestamp suffix in the concept name is used as a fallback. Each EE edge is oriented newer → older. We observed:

| Metric                  | Value  |
|-------------------------|--------|
| Observed earlier-share  | 0.807  |
| Shuffle baseline (mean) | ~0.499 |
| Shuffle baseline (std)  | ±0.083 |
| <i>z</i> -score         | ≈3.71  |

Table 1: Temporal bias in emergent→emergent link direction. The observed earlier-share substantially exceeds the shuffle baseline, indicating that newer emergent concepts preferentially link to older emergent concepts ( $p < 0.001$  under the shuffle null).

**Interpretation:** Emergent concepts in this run linked to older emergents substantially more than expected under random timing, consistent with a temporal “scaffolding” effect in which later concepts build upon earlier emergent structure. This is a structural property of the evolving concept graph, not a claim about consciousness or understanding.

### 6.3 Visual Evidence

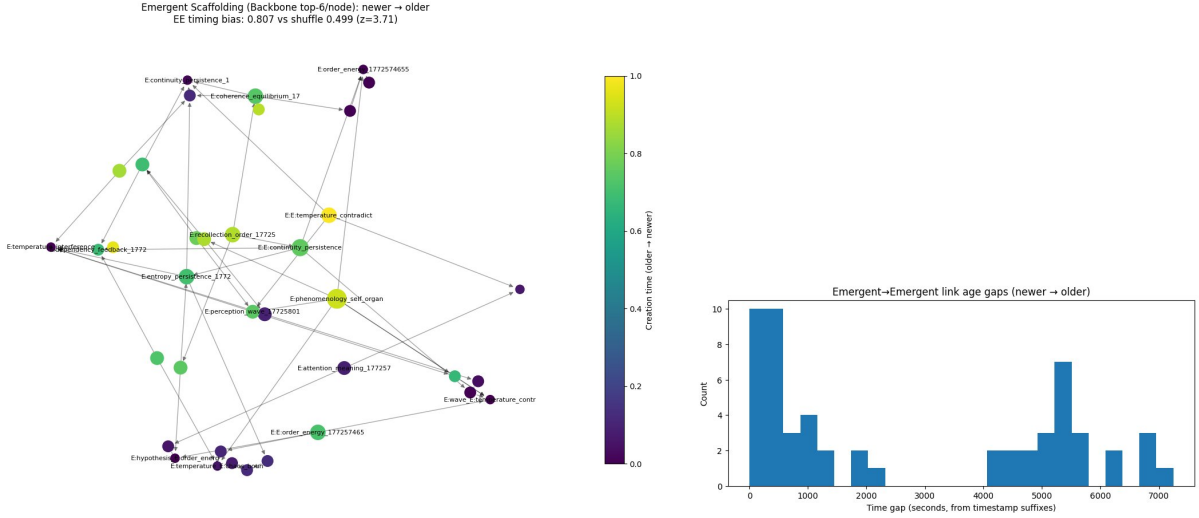


Figure 1: Emergent concept scaffolding structure from the representative 91-cycle cultivation run.

*Note:* To reproduce these figures, place the image files in a `figures/` directory as `emergent_scaffolding.png` and `link_age_gaps.png`.

## 7 Limitations

We note the following limitations to maintain defensibility:

1. **Language output:** The system’s language generation block is currently template-based. Output text quality does not reflect the sophistication of internal state dynamics.
2. **Timestamp ordering:** Temporal ordering of emergent concepts uses stored `creation_time` metadata as the primary source of truth, with the name-encoded timestamp suffix as a fallback. Future work should verify consistency between these two sources across all runs and flag any discrepancies.
3. **Single-run evidence:** The scaffolding results are drawn from a representative run. Multi-seed, multi-provider replication with distributional statistics (mean  $\pm$  sd) is needed before strong structural claims.
4. **No benchmark comparisons:** The architecture has not yet been evaluated against standardized cognitive architecture benchmarks.
5. **Dimensional constraints:** The current 5+5 dimensional ECWF space creates concept aliasing at scale. The v2 architecture (in progress) supports configurable dimensionality up to 64+64.
6. **Temperature heuristic:**  $T_g$  is derived from input complexity features, not from an energy functional. True thermodynamic grounding requires defining  $T = (\partial E / \partial S)^{-1}$  over a system energy functional.
7. **Dual coherence proxies:** The system currently maintains two distinct coherence measures—a wave-derived scalar (cognitive temperature) and a graph-derived density

proxy ( $\text{growth} \times \text{stability} \times \log N$ )—which are not yet formally disambiguated. Future work should assign distinct names and roles to each, and clarify which is appropriate for which analytical context.

8. **Ethical distance underspecification:** The global ethical distance  $\Delta_E$  in the pconnect edge policy is currently approximated from edge weight rather than computed from a stable system-wide ethical field. Formalizing this quantity is needed for the thermodynamic framing to be fully rigorous.

## 8 Related Work

Verdant-Minds shares structural elements with several existing frameworks while differing in its combination of wave-state dynamics with symbolic graph memory.

**Cognitive architectures:** SOAR [Laird(2012)] and ACT-R [Anderson et al.(2004)] provide modular cognitive pipelines with explicit memory systems. Verdant differs in its continuous wave-state representation and bidirectional bridge between symbolic and subsymbolic layers.

**Global Workspace Theory:** The CognitiveChunk flowing through governance-gated blocks resembles the global workspace paradigm [Baars(1988)], with the Three Kings playing the role of attentional gatekeepers.

**Conceptual Spaces:** The ECWF dimension space where concepts are mapped via embeddings is structurally a conceptual space [Gärdenfors(2000)], with emergent concept formation finding novel regions through wave dynamics.

**Integrated Information Theory:** The coherence geometry—particularly HCI and triangle validity checks—attempts something functionally similar to IIT’s  $\Phi$  measure [Tononi(2004)]: quantifying the degree to which a system’s state is integrated.

The key architectural novelty is the bidirectional bridge between a symbolic concept graph and a continuous wave function, with emergent concepts arising from wave dynamics and feeding back into the symbolic layer. This combination is not present in any of the above frameworks.

## 9 Roadmap

**Near-term:** Clean packaging (v2 extraction of ethomorphic core as separable package), stable chunked persistence, API server and dashboard for demonstration, benchmark harnesses.

**Medium-term:** Multiple-run statistical validation, controlled ablations (disable bridge, disable pconnect, vary entropy window), dimensional scaling experiments (16+16, 32+32, 64+64).

**Long-term:** Integration with external reasoning backends (LLM-powered language block), application to safety-sensitive domains requiring auditable cognition, and exploration of the ethomorphic core as a general-purpose module for embedding ethical geometry in other AI systems.

## 10 Conclusion

Verdant-Minds demonstrates a modular cognitive architecture with explicit memory, governance, and a wave-state core that yields measurable internal dynamics. Cultivation runs show not only emergent concept creation but evidence of temporal scaffolding among emergent concepts in the MemoryWeb backbone. This suggests the system is forming higher-order structure over time—an engineering property that can be instrumented, tested, and improved.

The separation of the ethomorphic core (wave engine, bridge, coherence geometry, ethical embedding) as an independent package establishes a foundation for applying the Ethomorphic AI principle to systems beyond the current Verdant architecture.

## A Reproducibility and Implementation Notes

### A.1 Environment and Entry Points

The cultivation runner is located at `scripts/verdant_llm_cultivator.py`. The core system initializes with a deterministic seed (default: 42) applied to NumPy and ECWF random state. Emergence is invoked from the continual learning block when wave magnitude exceeds 0.15, then filtered by the detector-level entropy window and magnitude threshold.

### A.2 Reproduction Commands

```
pip install -r requirements.txt
export GROQ_API_KEY=...
export GROQ_MODEL=llama-3.3-70b-versatile

# Fresh run
python scripts/verdant_llm_cultivator.py \
  --fresh --max-cycles 120 \
  --max-tokens-per-call 96 \
  --budget-mode light \
  --perturbation-interval 15

# Resume
python scripts/verdant_llm_cultivator.py \
  --resume

# Load explicit state
python scripts/verdant_llm_cultivator.py \
  --load-state \
  outputs/cultivation_state_<stamp>.json
```

### A.3 Output Artifacts

A single cultivation run produces timestamped outputs under `outputs/`:

- `cultivation_cycles_<stamp>.jsonl` — per-cycle records
- `cultivation_state_<stamp>.json` — final system state
- `cultivation_session_<stamp>.json` — session summary
- `significant_events_<stamp>.json` — major events

### A.4 Emergence Gating Parameters

| Gate               | Condition                                           | Default    |
|--------------------|-----------------------------------------------------|------------|
| Callsite magnitude | $\text{wave\_magnitude} > \text{threshold}$         | 0.15       |
| Entropy window     | $\text{sensitivity entropy} \in [lo, hi]$           | [0.3, 3.0] |
| Strength gate      | $\text{mean sensitivity} > \text{threshold}$        | 0.15       |
| Novelty gate       | <code>combo_key</code> unseen                       | —          |
| Surprise criterion | $\text{stability} \times \text{inverse similarity}$ | optional   |

Table 2: Emergence gating conditions and default parameter values.



## A.5 Determinism Notes

Even with a fixed core seed, strict run-to-run equivalence is not guaranteed due to: LLM provider nondeterminism, cultivator perturbation timing, and external API fallback chains. When reporting structural findings, we recommend distributional results across multiple runs (mean  $\pm$  sd) rather than single-run claims.

## A.6 Minimum Reproducibility Pack

For full reproducibility, attach: the JSONL cycle log, final state JSON, session summary, significant events log, the exact CLI command used, provider model string(s), and the git commit hash at run time.

## References

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